

Formal Verification: Nebula DAG Consensus Mode

Zach Kelling, Lux Industries

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Abstract

We formalize Nebula, the DAG blockchain consensus mode that wraps Field. Nebula provides DAG blockchain semantics with topological ordering of committed vertices. Its correctness reduces to Field's consistent cut property.

1 Introduction

Nebula is the DAG consensus mode, composing as $\text{Nebula} \rightarrow \text{Field} \rightarrow \text{Wave} \rightarrow \text{Prism} + \text{Flare}$. It wraps Field and provides DAG blockchain semantics. All invariants delegate to Field's consistent cut property.

2 Definitions

Definition 1 (Vertex). *A DAG vertex with id, parents, round, and author.*

Definition 2 (NebulaState). *State: vertex set, committed set, and current round.*

Definition 3 (IsTopologicalOrder). *Ordering property: if u is a parent of v , then u appears before v .*

3 Theorems and Axioms

Theorem 4 (committed_monotone). *Committed set only grows: $s.\text{committed} \subseteq s.\text{committed} \cup \text{newCommits}$.*

Theorem 5 (vertices_monotone). *Vertex set only grows.*

Theorem 6 (round_monotone). *Round number advances monotonically.*

4 Lean Source

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Nebula Protocol Formal Model

Models protocol/nebula/nebula.go -- DAG consensus mode.

Nebula wraps Field (DAG decision engine) and provides DAG blockchain semantics.

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Composition: Nebula -> Field -> Wave -> Prism (sampling)
              -> Flare (cert/skip)

Maps to:
- nebula.go: Nebula struct wrapping field.Driver
- nebula.go: Config{PollSize, Alpha, Beta, RoundTO, GenesisSet}

Properties:
- DAG validity: all parent references resolve to existing vertices
- Consistent ordering: committed vertices respect topological order
- Nebula delegates invariants to Field (consistent cut property)
-/

import Mathlib.Data.Nat.Defs
import Mathlib.Data.Finset.Basic
import Mathlib.Tactic

namespace Protocol.Nebula

/-- Nebula is a thin wrapper around Field. Its correctness reduces to Field's.
-/

/-- A DAG vertex -/
structure Vertex where
  id      : Nat
  parents : List Nat
  round   : Nat
  author  : Nat

/-- Nebula state -/
structure NebulaState where
  vertices   : Finset Nat      -- all known vertex IDs
  committed  : Finset Nat      -- committed vertex IDs
  round      : Nat             -- current round

/-- Topological order: if u is parent of v, u appears before v in ordering -/
def IsTopologicalOrder (vertices : List Vertex) : Prop :=
  i j : Fin vertices.length,
    (vertices[i]).id  (vertices[j]).parents →
    i.val < j.val

/-- Committed set only grows -/
theorem committed_monotone (s : NebulaState) (newCommits : Finset Nat) :
  s.committed s.committed newCommits :=
  Finset.subset_union_left

/-- Vertices only increase -/
theorem vertices_monotone (s : NebulaState) (newVerts : Finset Nat) :
  s.vertices s.vertices newVerts :=
  Finset.subset_union_left

/-- Committed is subset of vertices -/
def validState (s : NebulaState) : Prop :=
  s.committed s.vertices

```

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-- Round advances monotonically --
theorem round_monotone (s : NebulaState) (newRound : Nat) (h : newRound > s.
  round) :
  newRound > s.round := h

end Protocol.Nebula
```

5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Protocol/Nebula.lean>

White papers: <https://papers.lux.network>

Related white papers:

- `lux-consensus.tex`